

Q7		
(a)	The binding energy E of a nucleus is the energy required to separate the nucleus into its constituent free neutrons and protons. It is related to the mass defect Δm by the formula, $E = \Delta mc^2$.	
(b)(i)	The energy equivalent of $1.00\ u$ is given by, $= 1.00uc^2$ $= (1.66 \times 10^{-27})(3.00 \times 10^8)^2\ \text{J}$ $= \frac{(1.66 \times 10^{-27})(3.00 \times 10^8)^2}{(1.60 \times 10^{-19})(10^6)}\ \text{MeV}$ $= 933.75\ \text{MeV}$ $= 934\ \text{MeV}$	
(b)(ii)	For Nitrogen-14, Binding energy $E = (7(1.007276) + 7(1.008665) - 14.003074)uc^2$ $= 0.108513(934\ \text{MeV}) = 101.35\ \text{MeV}$ Binding energy per nucleon $= \frac{101.35}{14} = 7.24\ \text{MeV per nucleon}$	
(c)(i)	The energy released in the reaction $= \text{final binding energy} - \text{initial binding energy}$ $= 17(7.530) - [14(7.24) + 4(6.836)]$ $= 128.01 - 128.70$ $= -0.694\ \text{MeV}$ Hence there is actually a gain of mass in this reaction and 0.694 MeV of energy needs to be provided.	
(c)(ii)	We need at least 0.694 MeV of energy from the kinetic energy of the alpha particles as the oxygen nucleus and proton may also have kinetic energy after the reaction. Since the kinetic energy of the alpha particle is only 0.300 MeV, it is clearly insufficient and hence the reaction cannot proceed.	

